

Supplementary results for first hospitalisation counts before and during the COVID pandemic

The tables distinguish disjoint descriptive periods from overlapping model fits. Confidence intervals describe exploratory associations; no later-era effect is inferred by subtracting nested coefficients.

Descriptive periods

Supplementary Table S1. Monthly mean recorded counts in non overlapping calendar periods. Arithmetic means from aggregate period summaries.

Period	Months	CHD monthly mean	HF monthly mean
2013–2019	84	1315.3	252.0
2020–2022	36	926.3	172.6
2023	12	1026.9	191.3

Complete baseline panel

Supplementary Table S2. All twelve full window baseline weather models. Days-in-month offset, calendar-month indicators, four-degree-of-freedom natural time spline; Newey–West lag-six intervals. BH q-values cover these twelve Wald tests only.

Outcome	Exposure	Count ratio and 95% CI	p	BH q
CHD	Mean temperature / 1°C	0.993 (0.976–1.010)	0.424	0.726
CHD	Mean maximum / 1°C	0.993 (0.979–1.007)	0.321	0.642
CHD	Mean minimum / 1°C	0.994 (0.977–1.012)	0.509	0.764
CHD	Hot nights / 5 days	1.022 (1.002–1.042)	0.032	0.192
CHD	Very hot days / 5 days	0.999 (0.974–1.025)	0.962	0.962
CHD	Cold days / 5 days	0.995 (0.949–1.043)	0.823	0.900
HF	Mean temperature / 1°C	0.974 (0.947–1.002)	0.069	0.207
HF	Mean maximum / 1°C	0.981 (0.958–1.005)	0.113	0.272
HF	Mean minimum / 1°C	0.973 (0.947–1.000)	0.050	0.202
HF	Hot nights / 5 days	1.003 (0.976–1.031)	0.825	0.900
HF	Very hot days / 5 days	0.995 (0.963–1.028)	0.764	0.900
HF	Cold days / 5 days	1.073 (1.006–1.144)	0.031	0.192

The HF mean-minimum-temperature lag-six interval includes one on the unrounded scale; its upper bound is 1.0000496637 (Wald $p = 0.0504$). Rounding to 1.000 in Table S2 or 1.0000 in Table S4 does not change that interpretation.

The nested pre-2020 continuous-temperature fits were CHD mean temperature 0.980 (0.970–0.990), mean maximum 0.982 (0.974–0.991), and mean minimum 0.982 (0.972–0.992); HF corresponding fits were 0.945 (0.927–0.963), 0.958 (0.939–0.977), 0.944 (0.927–0.961). These estimates are not multiplicity-adjusted or a later-era contrast.

Calendar specification sensitivity

Supplementary Table S3. Complete official day panel under alternative calendar specifications. Ratios per five days with Newey–West lag-six intervals. Baseline and nested values are in main Table 2. Year indicators replace the spline. COVID-phase intercepts supplement the baseline in the full series; they are not a third window. Omission checks shorten the series. These sensitivity estimates have no multiplicity adjustment.

CHD

Specification	Hot nights	Very hot days	Cold days
Time spline 3 df	1.011 (0.990–1.032)	0.993 (0.967–1.020)	0.989 (0.934–1.046)
Time spline 6 df	1.024 (1.005–1.044)	0.999 (0.975–1.023)	0.999 (0.952–1.047)
Time spline 8 df	1.023 (1.005–1.042)	0.997 (0.974–1.021)	0.997 (0.950–1.047)
Year indicators	1.025 (1.000–1.050)	1.006 (0.983–1.029)	0.994 (0.939–1.052)
Omit first 12 months	1.020 (1.000–1.040)	0.995 (0.969–1.022)	1.008 (0.954–1.065)
Omit first 24 months	1.017 (0.999–1.036)	0.993 (0.966–1.020)	0.989 (0.930–1.053)
COVID phase intercepts	1.013 (0.994–1.033)	0.993 (0.973–1.013)	0.991 (0.943–1.041)

HF

Specification	Hot nights	Very hot days	Cold days
Time spline 3 df	0.993 (0.963–1.025)	0.989 (0.956–1.024)	1.066 (1.005–1.131)
Time spline 6 df	1.004 (0.977–1.032)	0.995 (0.963–1.028)	1.072 (1.005–1.143)
Time spline 8 df	1.002 (0.978–1.028)	0.993 (0.963–1.024)	1.062 (0.994–1.135)
Year indicators	1.004 (0.982–1.026)	0.997 (0.971–1.024)	1.061 (0.997–1.129)
Omit first 12 months	1.007 (0.977–1.038)	0.995 (0.963–1.029)	1.067 (0.995–1.143)
Omit first 24 months	1.003 (0.974–1.034)	0.988 (0.957–1.021)	1.043 (0.965–1.127)
COVID phase intercepts	0.994 (0.972–1.016)	0.990 (0.963–1.017)	1.074 (1.006–1.145)

Uncertainty constructions

Supplementary Table S4. Uncertainty intervals for all twelve baseline models. All four constructions use the same fitted coefficient. Lag-six q-values appear in Table S2 and do not apply to the other interval constructions. Four decimal places are retained for the near-null CHD hot-night lag-three bounds.

CHD

Exposure	Model CI	HC1 CI	NW3 CI	NW6 CI
Mean temperature / 1°C	0.9772–1.0092	0.9745–1.0120	0.9759–1.0105	0.9762–1.0102
Mean maximum / 1°C	0.9788–1.0073	0.9764–1.0098	0.9784–1.0077	0.9791–1.0069
Mean minimum / 1°C	0.9788–1.0098	0.9763–1.0124	0.9768–1.0119	0.9771–1.0116
Hot nights / 5 days	0.9950–1.0494	0.9973–1.0469	1.0003–1.0439	1.0019–1.0422
Very hot days / 5 days	0.9741–1.0254	0.9747–1.0247	0.9735–1.0260	0.9744–1.0251
Cold days / 5 days	0.9539–1.0371	0.9555–1.0353	0.9495–1.0419	0.9488–1.0427

HF

Exposure	Model CI	HC1 CI	NW3 CI	NW6 CI
Mean temperature / 1°C	0.9561–0.9929	0.9494–0.9999	0.9487–1.0006	0.9474–1.0020
Mean maximum / 1°C	0.9645–0.9981	0.9584–1.0044	0.9593–1.0035	0.9583–1.0046
Mean minimum / 1°C	0.9556–0.9909	0.9500–0.9967	0.9483–0.9986	0.9469–1.0000
Hot nights / 5 days	0.9702–1.0372	0.9706–1.0367	0.9739–1.0332	0.9759–1.0311
Very hot days / 5 days	0.9637–1.0274	0.9629–1.0282	0.9637–1.0273	0.9630–1.0281
Cold days / 5 days	1.0227–1.1253	1.0113–1.1381	1.0066–1.1434	1.0065–1.1435

Interpretation and source binding

The pre-2020 fit uses 84 of the 132 full-window months and a recalculated spline basis. No independent 2020–2023 coefficient, interaction test or weather standardisation is reported. General-population rates in older aggregate files are omitted because the population is not the eligible first-event risk set.

Tables draw from the repository files `outputs/tables/cvd_descriptive_annual_totals.csv`, `cvd_descriptive_covid_era_means.csv`, `cvd_core_robust_estimates.csv` and `cvd_trend_depletion_sensitivity.csv`. Baseline diagnostics are in `cvd_core_model_fit.csv`. All health values have HA_APPROVED_AGGREGATE provenance; no monthly governed files were read to produce this revision.

Visual comparison of nested and full windows

Supplementary Figure S1. Official day associations in overlapping analysis windows. Points and Newey–West lag-six 95% confidence intervals reproduce main Table 2. The pre-2020 84 months are contained within the full 132 months, and the spline basis is recalculated in each fit. The paired display is a sensitivity comparison, not independent pre/post estimates or a test of a change. Ratios are per five additional official days, without a consecutive-day requirement.

